

From Play to Dialogue: Designing a Cooperative Interactive Tabletop Game with Embedded Narrative and Discussion Prompts

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1 Introduction

Public discourse on socially complex issues like public surveillance usually occurs online or in formal institutional settings. Existing tools for civic and societal engagement can reach large audiences but fail to support face-to-face settings that allow people to think critically and make sense of complex issues together [Harris et al. 2026]. Serious games have become popular tools to raise awareness, but they prioritize learning outcomes rather than opportunities for collaborative discussions.

Narrative-centered multiplayer games suggest that collaborative sensemaking can be the central organizing principle of gameplay itself rather than just a measured outcome [Liang et al. 2025]. Research suggests that people develop deeper shared understanding when actively working through complex information together in person [Paul and Reddy 2010]. Interactive tabletop display around a shared space further suggests that face-to-face interaction is beneficial for collaborative communication and shared understanding [Maquil et al. 2024; Mateescu et al. 2019]. These findings point to cooperative interactive tabletop games as an underexplored space for meaningful civic discussion around public surveillance. We designed Co-op Watch, a cooperative interactive tabletop game featuring embedded narrative and choice-based discussion prompts, where players work together to remove surveillance devices spreading across their city before the community's trust and privacy collapses.

RQ: What interaction and discussion patterns emerge when embedded narrative with choice-based discussion prompts are integrated into a cooperative interactive tabletop display game about public surveillance?



Fig. 1. Co-op Watch deployed on a shared single-touch interactive tabletop display.

2 Related Work

Wong et al. (2023) found that exposure to surveillance scenarios surfaced values beyond privacy, including autonomy, safety, trust, and fairness; however, their findings came from individual interviews rather than collaborative group discussion. Embedded narrative is defined in prior work as a way of conveying background information through scripted narrative elements in digital interactive environments and games [Jenkins 2004]. Research on collaborative sensemaking is also well-documented in previous literature by Paul & Reddy (2010) as a social and interactive activity during which multiple individuals process ambiguity through sharing information and interpretations to build a collective understanding. Furthermore, research suggests that structured discussion mechanics can drive collaborative sensemaking among players, which helped inform our design decision of incorporating choice-based discussion prompts as embedded moments for group reflection during play [Liang et al. 2025]. However, there is still a need to examine how embedded narrative mechanics can shape cooperative group discourse around complex, values-based social topics.

3 Approach

To observe what interaction and discussion patterns emerge when embedded narrative and choice-based discussion prompts are integrated into cooperative gameplay around a socially complex topic, we designed Co-op Watch as both a game and a research method, using it as a vehicle for generating and capturing real social interactions as they emerge throughout gameplay. The game integrates narration of the surveillance topic into every layer of gameplay through board phase narrations, choice-based discussion prompts, and incident card events, so players cannot progress through the game without encountering and responding to the topic together.

Unlike prior research on interaction and discussion patterns in cooperative games, which has focused largely on digital platforms and remote gameplay, Co-op Watch is deployed on a shared interactive tabletop display where all players share the same information space simultaneously [Maquil et al. 2024]. Choice-based discussion prompts at the end of each board phase provide structured moments intended to elicit collaborative reflection within the flow of gameplay. The eight round progressive narrative escalates the ethical and social complexity of the surveillance topic across a single session, allowing us to observe how discussion and coordination patterns evolve among players as the narrative becomes more challenging over time. Together, the combination of cooperative game mechanics, progressive narrative, and embedded discussion prompts represents a novel integration not previously studied in interactive tabletop display games, offering a new context for observing interaction and discussion patterns around socially complex topics.

4 Method

The design process consisted of four main phases: literature review, game design, prototyping, and playtesting. The literature review helped inform the benefits of having a shared interactive tabletop display, cooperative board game design for facilitating discussion, and surveillance as a socially complex topic. Drawing upon these foundations, we developed the core game mechanics including rules, win/loss conditions, choice-based discussion prompts, player roles, and card mechanics.

For prototyping, we drew upon Pandemic's player-versus-board cooperative structure, in which players work together against an escalating system, a design pattern shown to elicit interdependence and communication among players, to mirror real-world scenarios of communities resisting surveillance expansion. [Leacock 2008; Seif El-Nasr et al. 2010]. We sketched initial board layouts, developed the game narrative, and transferred the physical prototype into a digital version built in Claude Code, which was tested on the touchscreen display. Pixel art was the concept chosen to make the game more approachable and reduce potential barriers on engaging with a sensitive topic. Multiple peer playtesting sessions were conducted to evaluate game mechanics, player engagement, and accessibility of the game.

This study is pending Institutional Review Board (IRB) approval. Participants ages 18-35 will be recruited via snowball sampling for a single-session study lasting approximately 60-90 minutes, with sessions being video and audio recorded. Data will be collected through observation of 2-4 participants playing on the shared single-touch interactive tabletop display across eight narrative rounds, each concluding with an embedded choice-based discussion prompt designed to elicit group deliberation. After gameplay, participants will complete a post-game individual survey measuring perceived collaboration quality, social dynamics, and narrative engagement during the session. A structured closing group debrief will then be facilitated to capture participants' immediate collective reflections on the experience. Observation notes will be recorded using the Quadripartite Taxonomy framework, focusing on WHAT game element is being discussed and HOW players are engaging collaboratively to capture coordination, cooperation, and reflective communication patterns, along with whether embedded narrative enters players discussion and the depth of collaboration engagement [Baykal et al. 2023].

5 Pilot Observations

Pilot playtests were conducted on an earlier version of Co-op Watch that did not yet include embedded narrative or choice-based discussion prompts. While players engaged with cooperative mechanics, discussion around the surveillance topic did not emerge during gameplay. Players focused primarily on coordinating game actions rather than reflecting on social issues embedded in the game. One player adopted a verbal direction role by coordinating other players' moves, which is consistent with prior literature on single-touch shared displays [Clayphan et al. 2018], suggesting that single-touch constraints may shape how players divide roles regardless of task context. The same player took on an informal "explainer" role, helping other players understand the game mechanics during gameplay.

Additionally, players also explicitly requested more storytelling and structure to give the game narrative progression.

6 Limitations

This study has several limitations. First, pilot playtesting was conducted on an earlier version of Co-op Watch that did not yet include the embedded narrative and choice-based discussion prompts which are central to the research question. As a result, pilot observations reflect behaviors from incomplete design conditions that may not fully represent how players will interact with the current version. Second, pilot participants were recruited from a university setting, which may not reflect how the game would perform with diverse community groups. Third, for the planned formal study, sample size and convenience sampling through snowball recruitment may limit generalizability. Fourth, for the planned formal study, the presence of study investigators and recording equipment during sessions may influence how participants engage with and discuss the game. Fifth, this study focuses on interaction and discussion patterns that emerge when embedded narrative with choice-based discussion prompts, so future comparative work could examine the same game to address questions of causation with and without embedded narrative or choice-based discussion prompts.

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